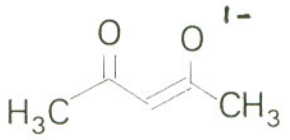
# INORGANIC CUMULATIVE EXAM

October 24, 1992

Answer 3 of the following questions:

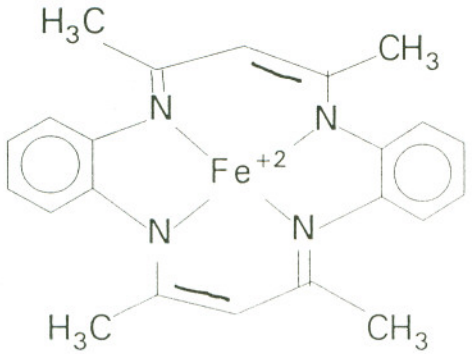
1) The ion $CuCl_6^{4-}$ can be precipitated from concentrated HCl with the appropriate choice of amine cation. Make a prediction of the observed Cu - Cl bond distances for this ion. How would you anticipate the experimental values would compare to those for $CuCl_4^{1-}$ ? [The tabulated values of the covalent radii of Cu and Cl are 1.38 and 0.99 angstrom respectively]

2) The complex Pt(acac)Cl(cis-2-butene) shows a single methyl resonance for the olefin $CH_3$ groups at room temperature but two different resonances below -25 $^{o}C$. Account for this observation. acac = acetylacetonate anion




3) One isomer of $Pt(NH_3)_2Cl_2$ reacts with pyridine to give a single isomer of $[Pt(NH_3)_2(py)_2]Cl_2$ which reacts with HCl to give $[Pt(NH_3)(py)Cl_2]$. The second isomer of $[Pt(NH_3)_2(Cl)_2]$ reacts with pyridine to give a different isomer of $[Pt(NH_3)_2(py)_2]Cl_2$ which reacts with HCl to give a mixture of one isomer of $[Pt(NH_3)_2Cl_2]$ and the same geometric isomer of $[Pt(py)_2Cl_2]$. Assign the geometries of all the isomers and account for these observations.

4) Suggest a synthesis for the the compound below which does not involve reaction of the pre-formed macrocyclic ligand with Fe(II) cation.




Note : the macrocylic ligand is essentially planar and there are two water ligands axial to the plane.